TOPIC

TASK

PREVIOUS MODEL

GRAPH MODEL

IMPLEMENTATION

Case Study #4: Research Project Management Office (RMPO)

13 November 2025

Jirapat Datephanyawat (st125843)

Paradorn Khanongsuwan (st125894)

Thet Htun Swe (st126391)

Prepared by group Data Group #2

Agenda

Kookle University
Project Initiation Form

Date: 1 Jan 2019
Project Title: Ontology-based Plant Disease Detection System
Project Description: The project aims to ...

ywords: Ontology, Plant Disease, Intelligent System

oject Category:
- Research and Development
- Training/Capacity Building
- Conference/Workshop

search Areas of the Project: Semantic Web, Ontology, Knowledge Representation
oject Duration: 12 months
Contract Period: 1 Jan 2019 – 31 Dec 2019
Budget: 1,000,000 THB
Installment:

No.	Date	%Budget	Amount
1	30 April 2019	30	300,000
2	31 August 2019	30	300,000
3	31 December 2019	40	400,000

Principle Investigator:
1. Name: Mark Zuckerberg Email: mark@kmail.com
   Department ICT Department Area of Expertise: Social Analytics, AI
2. Name: Elon Musk Email: musk@kmail.com
   Department ICT Department Area of Expertise: Knowledge Representation

Project Member:
1. Name: Tim Berners Lee Email: tim@kmail.com
   Department ICT Department Area of Expertise: Semantic Web Ontology
2. Name: Email:
   Department Area of Expertise:

Donor/Sponsor Name: Thailand Research Fund
Donor Type:
- Research Organization
- International Organization
- Private Company

Donor Contact Info: Mr. Somchai N. CMMU Building Phayathai Bangkok 10400 Thailand
Tel: (+66) 2 343 1500

Student Assistant:
1. Name: Nina Billa Email: nina@kmail.com
   Department: ICT
   Responsibility: Programming
   Maximum working hours per month: 40 Rate per hour: 300
   Start Date: 1 Feb 2019 End date: 30 June 2019
2. Name: Lucy Dell Email: lucy@kmail.com
   Department: Business Management
   Responsibility: Project Coordinator
   Maximum working hours per month: 40 Rate per hour: 300
   Start Date: 1 Jan 2019 End date: 31 Dec 2019

Main Focus of the Case Study

This case study focuses on the management of research project data at Kookle University. The RMPO (Research Project Management Office) is responsible for maintaining and tracking information about **research projects, grants, donors, and the faculty, staff, and students** involved.

OBJECTIVE

1.Maintain Data with CRUD Method

2.Analyze and report statistics for management

3.Tracking, planning and forecasting

4. Research performance of each faculty

Spread sheet

Business Rules, Data Maintained and Constraints

Data Maintained

- **Personal details:** Name, Email, Contact, Gender
- **Department info:** Department name, Domain
- **Project details:** Title, Status, Duration, Dates
- **Work details:** Roles, Responsibilities, Working Hours, Hourly Rate
- **Financial info:** Donor, Donation Amount, Installments, Due Dates
- **Classification:** Category, Keywords

Business Rules

- Every project must have staff and domain assigned
- Students assist projects within project duration
- Each person and student belongs to one department at a time
- Donations are linked to one donor and one project
- Installment percentages must total 100%
- Project start date must precede end date

Data Constraints

- **Unique identifiers** for Person, Project, Donor, Department
- **Mandatory fields:** names, titles, amounts, and dates
- **Valid relationships:** all connections must link existing nodes
- **Start date < End date** for all time-based data
- **Installment** totals = **donation** total
- **One category** per project; at least **one domain** required

TOPIC

TASK

PREVIOUS MODEL

GRAPH MODEL

IMPLEMENTATION

Previous model

TOPIC

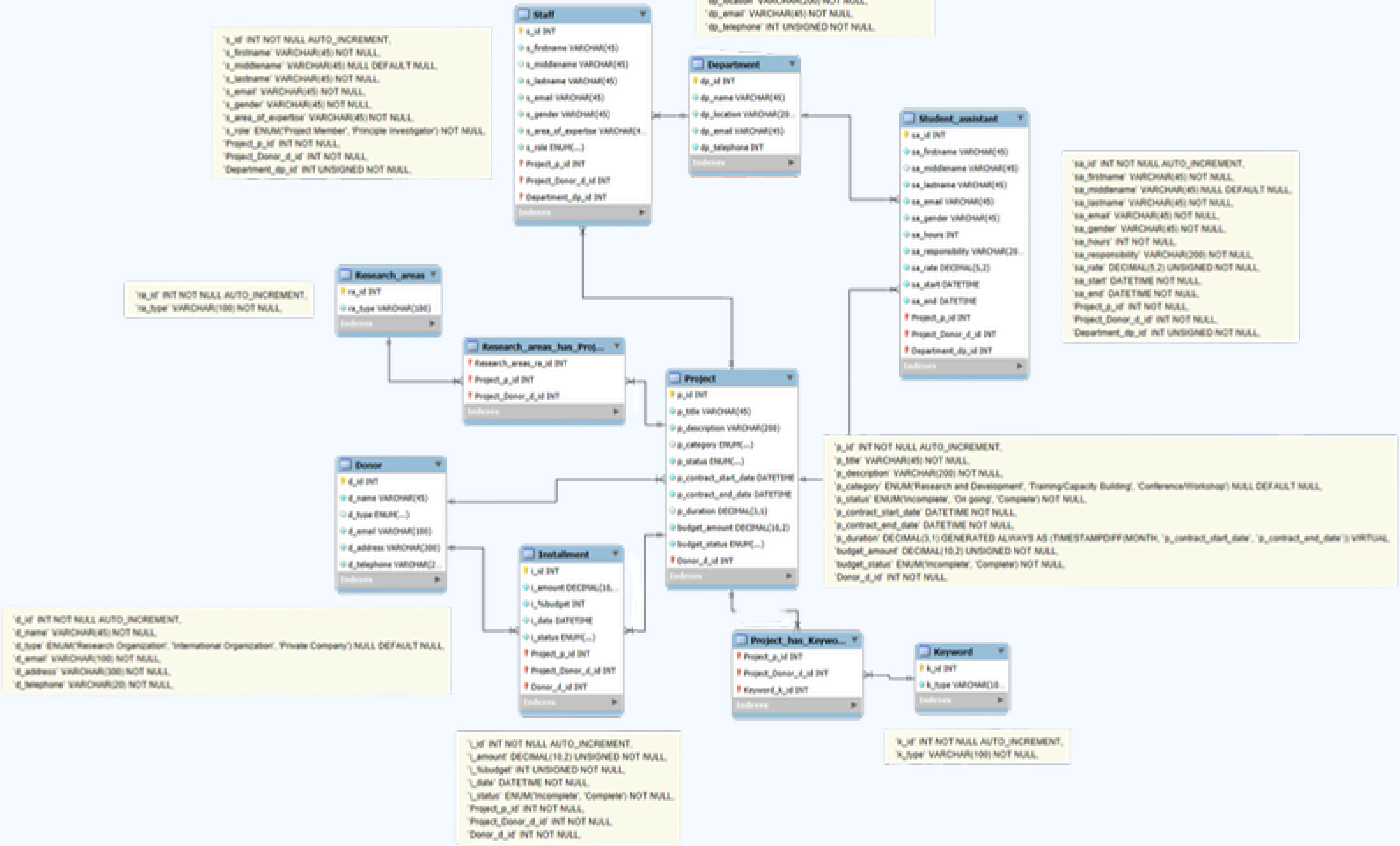
TASK

PREVIOUS MODEL

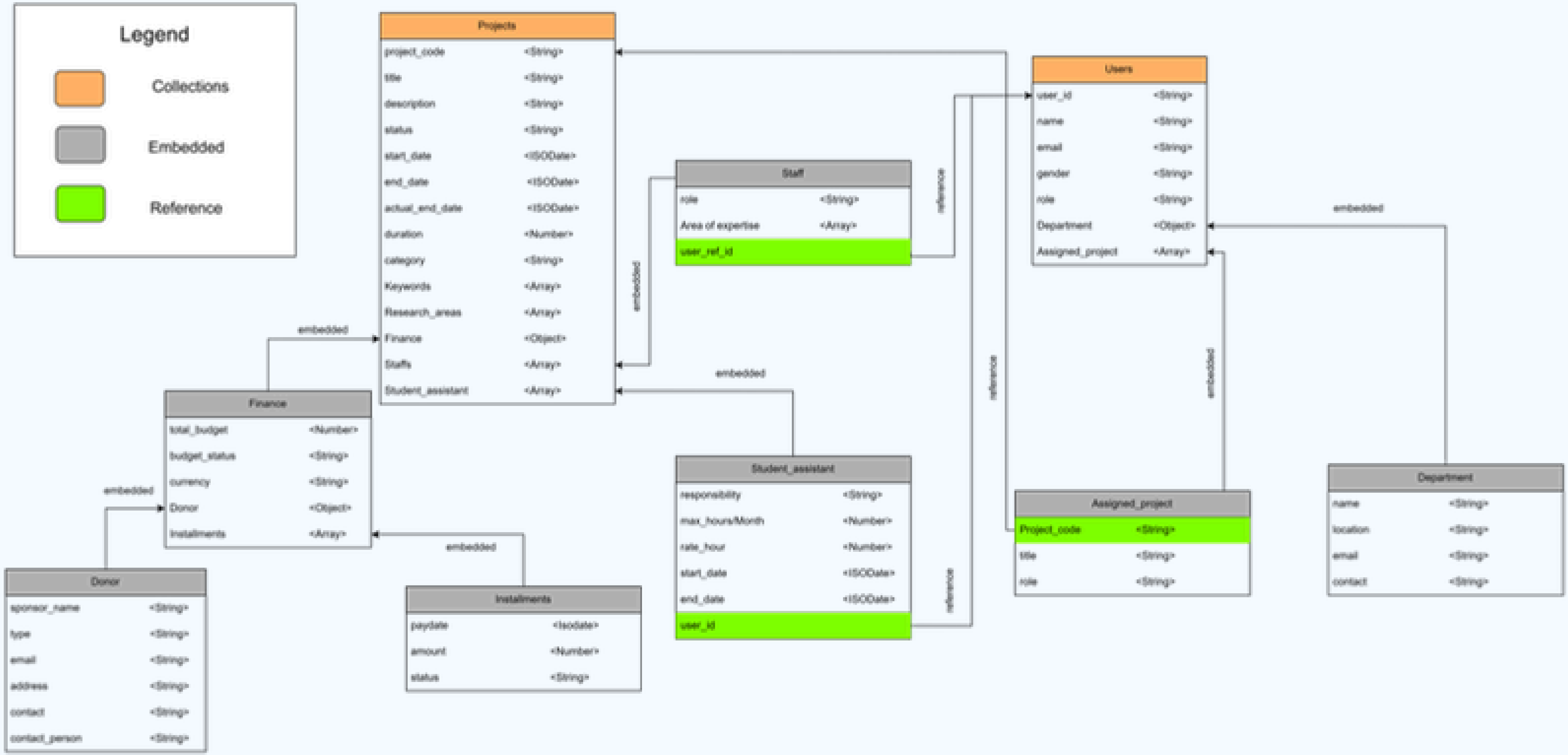
GRAPH MODEL

IMPLEMENTATION

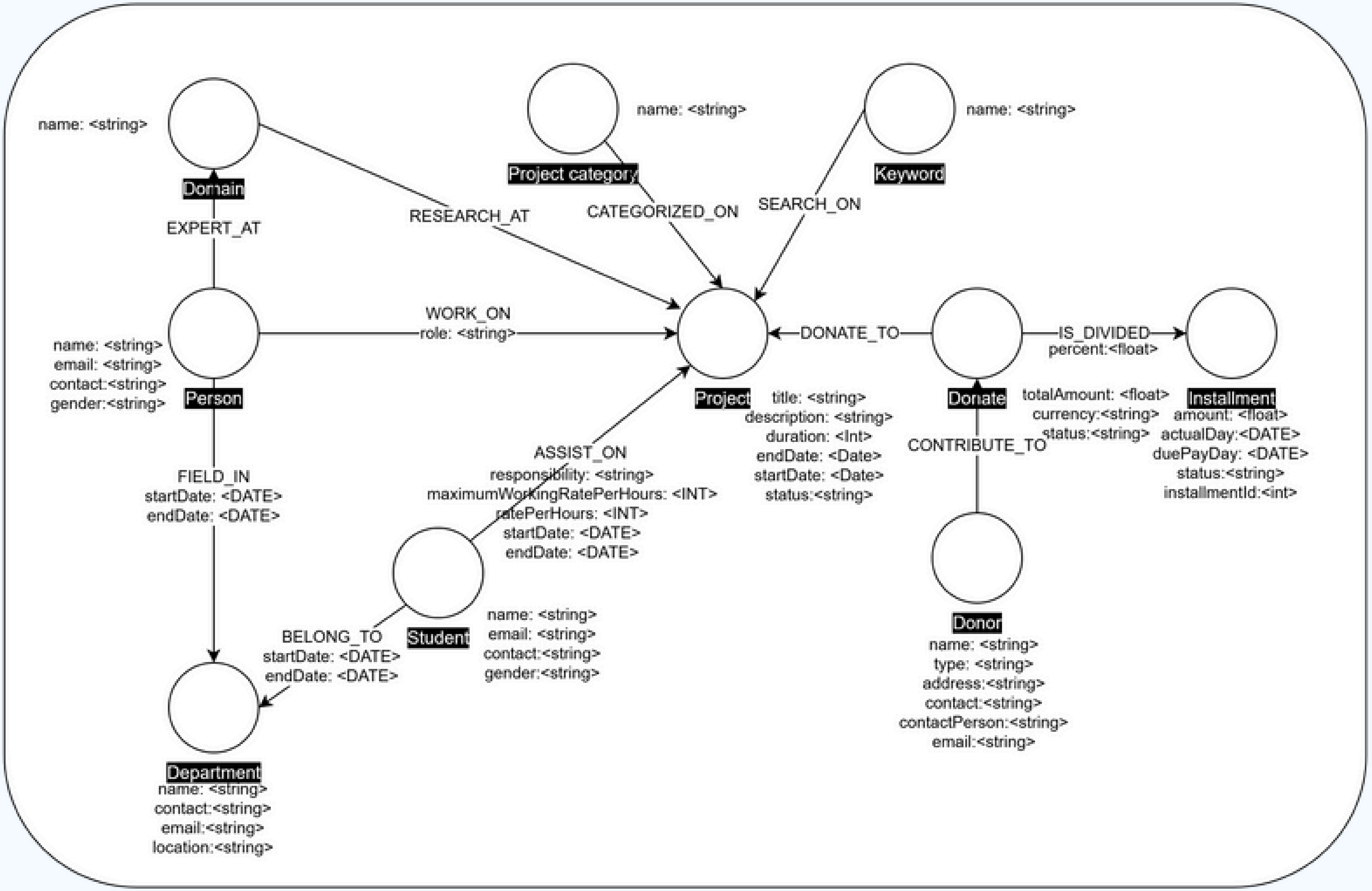
ER Diagram



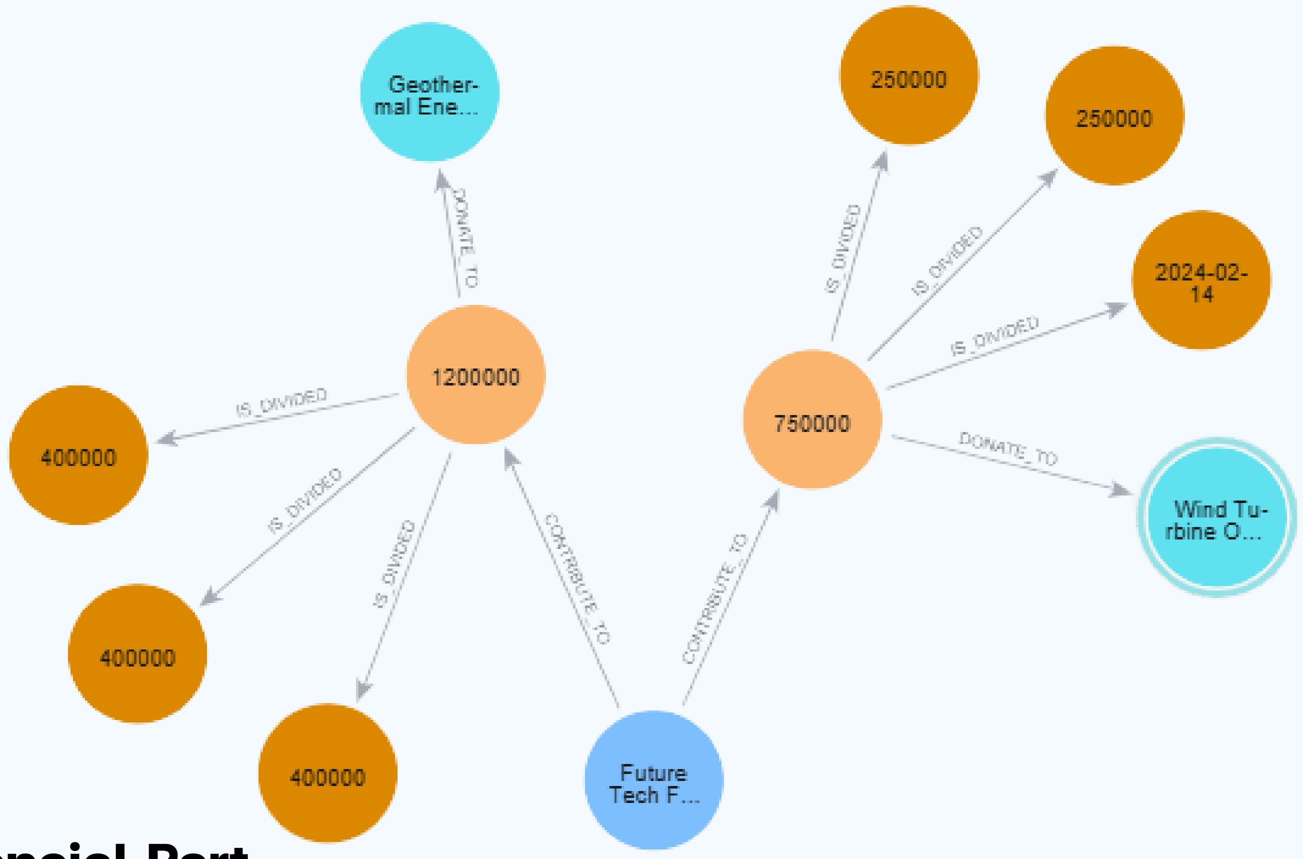
Document Diagram

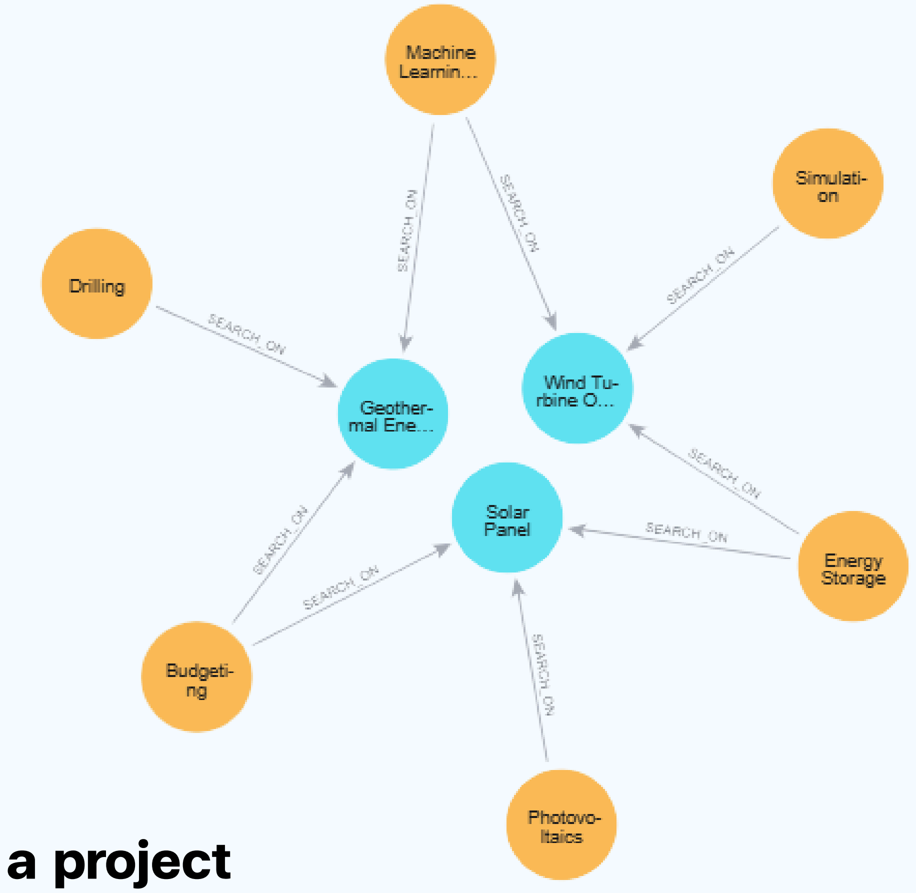


Conceptual Diagram



Refactoring





Financial Part

Graph Normalization, Instead of recording financial details on the project or relationships (risking data redundancy), we promoted the transaction details to their own nodes. We separated the total donation amount into a central (Donate) Node and its scheduled payments into dedicated (Installment) Nodes linked via [:IS_DIVIDED].

Impact

This enables Cash Flow Analysis and Financial Granularity (e.g., calculating completion percentage based on payments received) while eliminating data integrity risks. Crucially, it allows us to directly query for the largest financial figures—whether it's the totalAmount on the Donate node or the highest single amount on an Installment node—without complex filtering.

Searching a project

We promoted attributes like Keywords, Domains, and Staff from simple list properties inside the project node to independent, dedicated Nodes ((Keyword), (Domain), (Person)). This creates a web of interconnected knowledge.

Impact

This is significantly faster and computationally less demanding than scanning string properties. It enables Network Analysis & Expertise Mapping by allowing queries to efficiently start and traverse directly from the subject of expertise (e.g., finding the top-funded research domain).

Implementation

TOPIC

TASK

PREVIOUS MODEL

GRAPH MODEL

IMPLEMENTATION

Show project timeline and payment schedule

```
MATCH (p:Project)<-[:DONATE_TO]-(d:Donate)-[:IS_DIVIDED]->(i:Installment)
RETURN
 p.title  AS  Project,
 p.startDate  AS  ProjectStart,
 p.endDate  AS  ProjectEnd,
 i.installmentId  AS  InstallmentID,
 i.duePayDay  AS  PaymentDueDate,
 i.amount  AS  PaymentAmount,
 i.status  AS  PaymentStatus
ORDER BY
 Project, PaymentDueDate
```

RESULT

	Project	ProjectStart	ProjectEnd	InstallmentID	PaymentDueDate	PaymentAmount	PaymentStatus
1	"Geothermal Energy Plant"	2024-06-01	2027-06-01	"I-003"	2024-06-15	400000	"Pending"
2	"Geothermal Energy Plant"	2024-06-01	2027-06-01	"I-008"	2025-01-15	400000	"Pending"
3	"Geothermal Energy Plant"	2024-06-01	2027-06-01	"I-009"	2025-06-15	400000	"Pending"
4	"Solar Panel"	2023-01-15	2025-01-14	"I-001"	2023-03-01	100000	"Paid"
5	"Solar Panel"	2023-01-15	2025-01-14	"I-004"	2023-09-01	200000	"Paid"
6	"Solar Panel"	2023-01-15	2025-01-14	"I-005"	2024-03-01	200000	"Pending"
7	"Wind Turbine Optimization"	2024-01-01	2026-01-01	"I-002"	2024-02-15	250000	"Paid"
8	"Wind Turbine O	2024-01-01	2026-01-01	"I-006"	2024-08-15	250000	"Pending"

Calculate total budget amount by research area for current year

```
WITH
date({year: 2025, month: 1, day: 1}) AS startOfYear,
date({year: 2025, month: 12, day: 31}) AS endOfYear

MATCH (d:Domain)-[:RESEARCH_AT]->(p:Project)<-[:DONATE_TO]-(don:Donate)
WHERE p.startDate <= endOfYear AND p.endDate >= startOfYear

RETURN
 d.name AS ResearchArea,
 SUM(don.totalAmount) AS TotalBudget,
 don.currency AS Currency
ORDER BY TotalBudget DESC
```

RESULT

	ResearchArea	TotalBudget	Currency
1	"Renewable Energy"	1700000	"THB"
2	"Data Science"	1250000	"THB"
3	"Geology"	1200000	"THB"
4	"Aerodynamics"	750000	"THB"
5	"Material Science"	750000	"THB"

TOPIC

TASK

PREVIOUS MODEL

GRAPH MODEL

IMPLEMENTATION

Identify the areas of expertise with the fewest staff members in ongoing projects, for future recruitment planning

```
MATCH (p:Person)-[:WORK_ON]->(proj:Project)
WHERE proj.endDate > date()
WITH collect(DISTINCT p) AS activeStaff
MATCH (d:Domain)

OPTIONAL MATCH (d)<-[:EXPERT_AT]-(p:Person)
WHERE p IN activeStaff
RETURN d.name AS Domain, count(p) AS ActiveStaffCount
ORDER BY ActiveStaffCount ASC
```

RESULT

	Domain	ActiveStaffCount
1	"Renewable Energy"	1
2	"Aerodynamics"	1
3	"Material Science"	1
4	"Data Science"	2
5	"Geology"	2

Identify top 5 research areas by total funding received

```
MATCH (d:Domain)-[:RESEARCH_AT]->(p:Project)<-[:DONATE_TO]-(don:Donate)
RETURN
 d.name AS ResearchArea,
 SUM(don.totalAmount) AS TotalFunding,
 don.currency AS Currency
ORDER BY
 TotalFunding DESC
LIMIT 5
```

RESULT

	ResearchArea	TotalFunding	Currency
1	"Renewable Energy"	1700000	"THB"
2	"Data Science"	1250000	"THB"
3	"Geology"	1200000	"THB"
4	"Aerodynamics"	750000	"THB"
5	"Material Science"	750000	"THB"

Calculate total funding received from each donor type by year

```
MATCH (d:Donor)-[:CONTRIBUTE_TO]->(don:Donate)-[:IS_DIVIDED]->(i:Installment)
WHERE i.actualDay IS NOT NULL
RETURN
 d.type AS DonorType,
 i.actualDay.year AS Year,
 don.currency AS Currency,
 SUM(i.amount) AS TotalReceived
ORDER BY
 DonorType, Year
```

RESULT

	DonorType	Year	Currency	TotalReceived
1	"International Organization"	2023	"THB"	300000
2	"Private Company"	2024	"THB"	250000

TOPIC

TASK

PREVIOUS MODEL

GRAPH MODEL

IMPLEMENTATION

Track project keyword trends over the past 5 years

```
MATCH (k:Keyword)-[:SEARCH_ON]->(p:Project)
WHERE p.startDate >= date() - duration('P5Y') AND p.startDate <= date()
RETURN k.name AS Keyword, p.startDate.year AS Year, count(p) AS ProjectCount
ORDER BY Keyword, Year
```

RESULT

	Keyword	Year	ProjectCount
1	"Budgeting"	2023	1
2	"Budgeting"	2024	1
3	"Drilling"	2024	1
4	"Energy Storage"	2023	1
5	"Energy Storage"	2024	1
6	"Machine Learning"	2024	2

Comparative Analysis of the Three Models

FEATURE	1. SQL (MYSQL)	2. DOCUMENT (MONGODB)	3. GRAPH (NEO4J)
Core Structure	Tables	Documents	Nodes & Relationships
Data Storage Method	Data is **normalized** (split) into many distinct tables (e.g., `Project`, `Person`, `Donor`).	Related data is **embedded** into a single, nested main document (e.g., `Staff` and `Financials` are *inside* the `Projects` document).	Data is stored as nodes (`Project`, `Person`). The **relationship itself** is a primary entity that directly links the nodes.
Handling Relationships	Uses **Foreign Keys (FKs)** to reference the ID of another table (e.g., `project_id`, `person_id`).	**1. Embedding:** Stores `Person` data (Staff) *inside* the `Project` document. **2. Referencing:** Stores `user_id` s in an array, requiring an application-level `JOIN` (or `$lookup`).	**Relationships** are first-class citizens (`[:LEAD_BY]`, `[:MEMBER_OF]`). They are stored as direct, physical connections (pointers).
Querying Method	Requires many `JOIN` operations to combine data from different tables. *(e.g., `Project` JOIN `Staff_Junction` JOIN `Person`)*	**Easy query** if fetching one document's data. **Hard query** if searching *across* documents (e.g., "Which projects is Alice in?").	Uses **Pattern Matching (Traversal)** to "walk" the relationships. *(e.g., `(Person {name: 'Alice'})-[:MEMBER_OF]->(Project)`)*
Data Redundancy	**Very Low.** Data is normalized. 'Alice' is stored only once in the `Person` table.	**High (if embedding).** 'Alice's' name might be stored in *every* project she's in. A name change requires multiple updates.	**Very Low.** The `Person` node for 'Alice' exists only once. Projects just "point" a relationship to that single node.
Flexibility (Schema)	**Low (Schema-on-Write).** A rigid, pre-defined schema is required before data is added.	**High (Schema-on-Read).** Each document can have a different structure. New fields can be added easily.	**High (Semi-Structured).** Flexible like a Document (add properties easily) but can enforce data integrity with constraints (like SQL).

TOPIC
TASK
PREVIOUS MODEL
GRAPH MODEL
IMPLEMENTATION

CONCLUSION

- The **Neo4j graph model** efficiently represents the complex relationships between projects, donors, departments, domains, and people within the Research Project Management Office (RMPO).
- Through graph normalization, the model eliminates redundancy and supports fast, intuitive querying for research analysis and financial tracking.
- Implementing dedicated nodes for **Donate** and **Installment** enables accurate financial monitoring, while relationships such as **WORK_ON**, **ASSIST_ON**, and **RESEARCH_AT** reveal staff expertise and collaboration patterns.
- The analytical Cypher queries demonstrate that Neo4j can provide real-time insights such as funding trends, research domain performance, and staffing distribution that are difficult to achieve in traditional relational systems.

Overall, this project highlights Neo4j's strength in **data interconnectivity, performance, and scalability**, making it a powerful solution for managing and analyzing research project ecosystems.

Thank you